BIOL 300 Formulae and Statistical Tables

Please do not mark the formula sheets or tables.

$$\bar{Y} = \frac{\sum_{i=1}^{n} Y_i}{n}$$

$$s = \sqrt{\frac{\sum(Y_i - \bar{Y})^2}{n-1}} = \sqrt{\frac{\sum(Y_i^2) - n\bar{Y}^2}{n-1}}$$

$$SE_{\bar{Y}} = \frac{s}{\sqrt{n}}$$

$$\Pr[A|B] = \frac{\Pr[B|A]\Pr[A]}{\Pr[B]}$$

$$\Pr[x] = \binom{n}{x} p^x (1-p)^{n-x}$$

$$\binom{n}{x} = \frac{n!}{x!\,(n-x)!}$$

$$p' = \frac{X + 2}{n + 4}$$

$$\left(p' - 1.96\sqrt{\frac{p'(1-p')}{n+4}}\right) < p < \left(p' + 1.96\sqrt{\frac{p'(1-p')}{n+4}}\right)$$

$$\chi^2 = \sum_i \frac{(Observed_i - Expected_i)^2}{Expected_i}$$

$$\widehat{OR} = \frac{ad}{bc}$$

$$\ln(\widehat{OR}) - Z * SE[\ln(\widehat{OR})] \leq \ln(OR) \leq \ln(\widehat{OR}) + Z * SE[\ln(\widehat{OR})]$$

$$P[x] = \frac{\mu^x e^{-\mu}}{x!}$$

$$f[x] = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$Z = \frac{x - \mu}{\sigma}$$

$$\frac{df\, s^2}{\chi^2_{\frac{\alpha}{2},df}} \leq \sigma^2 \leq \frac{df\, s^2}{\chi^2_{1-\frac{\alpha}{2},df}}$$

$$\bar{Y} \pm SE_{\bar{Y}} t_{\alpha(2),df}$$

$$t = \frac{\bar{Y} - \mu}{s/\sqrt{n}}$$

$$t = \frac{\bar{d} - \mu_{d0}}{SE_d}$$

$$t = \frac{(\bar{Y}_1 - \bar{Y}_2) - (\mu_1 - \mu_2)_0}{SE_{\bar{Y}_1 - \bar{Y}_2}}$$

$$(\bar{Y}_1 - \bar{Y}_2) \pm SE_{\bar{Y}_1 - \bar{Y}_2}\, t_{\alpha(2),df}$$

$$SE_{\bar{Y}_1 - \bar{Y}_2} = \sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$s_p^2 = \frac{df_1 s_1^2 + df_2 s_2^2}{df_1 + df_2}$$

$$t = \frac{\bar{Y}_1 - \bar{Y}_2 - (\mu_1 - \mu_2)_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\left(\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}\right)}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 - U_1$$

Please do not mark the formula sheets or tables.

$$F = \frac{MS_{groups}}{MS_{error}} \qquad \bar{Y} = \frac{\sum n_i(\bar{Y}_i)}{N}$$

$$MS_{error} = s^2_{pooled} = \frac{\sum s_i^2(n_i - 1)}{N - k}$$

$$MS_{groups} = \frac{\sum n_i(\bar{Y}_i - \bar{Y})^2}{k - 1}$$

$$R^2 = \frac{SS_{groups}}{SS_{total}}$$

$$q = \frac{\bar{Y}_i - \bar{Y}_j}{SE} \quad SE = \sqrt{s^2_{pooled}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

$$r = \frac{\left(\sum_i (X_i - \bar{X})(Y_i - \bar{Y})\right)}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}$$

$$t = \frac{r}{SE_r}$$

$$SE_r = \sqrt{\frac{1-r^2}{n-2}}$$

$$z = 0.5 \ln\left(\frac{1+r}{1-r}\right)$$

$$\sigma_z = \sqrt{\frac{1}{n-3}} \qquad r_s = 1 - \frac{6\sum d_i^2}{n^3 - n}$$

$$\sum (X - \bar{X})(Y - \bar{Y}) = \sum XY - \frac{(\sum X)(\sum Y)}{n}$$

$$\sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$b = \frac{\left(\sum_i (X_i - \bar{X})(Y_i - \bar{Y})\right)}{\sum_i (X_i - \bar{X})^2}$$

$$a = \bar{Y} - b\bar{X}$$

$$SS_{Total} = \sum Y_i^2 - \frac{(\sum Y_i)^2}{n}$$

$$SS_{regression} = b \sum (X_i - \bar{X})(Y_i - \bar{Y})$$

$$SS_{residual} = SS_{Total} - SS_{regression}$$

$$MS_x = \frac{SS_x}{df_x}$$

$$r^2 = \frac{SS_{regression}}{SS_{total}}$$

$$SE_b = \sqrt{\frac{MS_{residual}}{\sum (X_i - \bar{X})^2}}$$

$$MS_{residual} = \frac{\sum (Y_i - \bar{Y})^2 - b\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 2}$$

$$b \pm t_{(\alpha(2),df)} SE_b$$

$$\hat{Y} \pm t_{(\alpha(2),df)} SE_{\hat{Y}}$$

$$t = \frac{b - \beta_0}{SE_b}$$

$$(MS_{error})_p = \frac{(SS_{error})_1 + (SS_{error})_2}{(DF_{error})_1 + (DF_{error})_2}$$

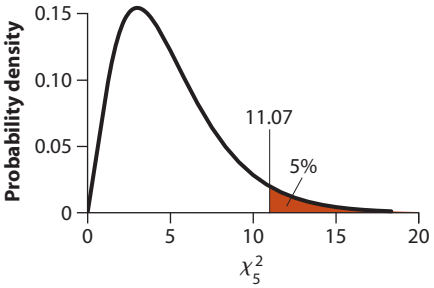
$$SE_{b_1 - b_2} = \sqrt{\left(\frac{(MS_{error})_p}{(\sum (X - \bar{X})^2)_1} + \frac{(MS_{error})_p}{(\sum (X - \bar{X})^2)_2}\right)}$$

$$L[value|data] = \Pr[data|parameter = value]$$

$$G = 2ln\left(\frac{L[maximum\ likelihood\ value\ of\ parameter|data]}{L[parameter\ value\ under\ H_0|data]}\right)$$

Statistical Table A: The χ^2 distribution

This table gives critical values of the χ^2 distribution. The critical value $\chi^2_{df,\alpha}$ defines the area α in the right tail of the χ^2 distribution with df degrees of freedom. To find a critical value in the table, select the desired α along the top row and the number of degrees of freedom df in the far left column. For example, if $df = 5$ and $\alpha = 0.05$, then the critical value is 11.07; that is, the probability of a value greater than or equal to 11.07 is 0.05.



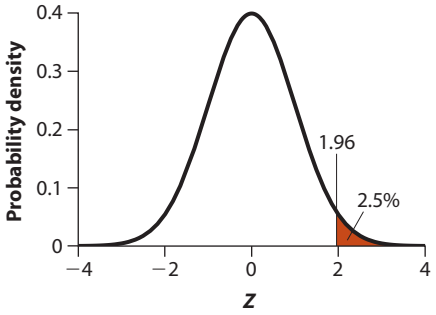
The critical value for a χ^2 distribution having $df = 5$ when $\alpha = 0.05$. The area in red indicates the tail probability corresponding to 0.05, or 5%. The boundary of the red section is $\chi^2 = 11.07$. The probability of a value greater than or equal to 11.07 is 0.05.

	α									
df	**0.999**	**0.995**	**0.99**	**0.975**	**0.95**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**
1	0.0000016	0.000039	0.00016	0.00098	0.00393	3.84	5.02	6.63	7.88	10.83
2	0.002	0.01	0.02	0.05	0.10	5.99	7.38	9.21	10.60	13.82
3	0.02	0.07	0.11	0.22	0.35	7.81	9.35	11.34	12.84	16.27
4	0.09	0.21	0.30	0.48	0.71	9.49	11.14	13.28	14.86	18.47
5	0.21	0.41	0.55	0.83	1.15	11.07	12.83	15.09	16.75	20.52
6	0.38	0.68	0.87	1.24	1.64	12.59	14.45	16.81	18.55	22.46
7	0.60	0.99	1.24	1.69	2.17	14.07	16.01	18.48	20.28	24.32
8	0.86	1.34	1.65	2.18	2.73	15.51	17.53	20.09	21.95	26.12
9	1.15	1.73	2.09	2.70	3.33	16.92	19.02	21.67	23.59	27.88
10	1.48	2.16	2.56	3.25	3.94	18.31	20.48	23.21	25.19	29.59
11	1.83	2.60	3.05	3.82	4.57	19.68	21.92	24.72	26.76	31.26
12	2.21	3.07	3.57	4.40	5.23	21.03	23.34	26.22	28.30	32.91
13	2.62	3.57	4.11	5.01	5.89	22.36	24.74	27.69	29.82	34.53
14	3.04	4.07	4.66	5.63	6.57	23.68	26.12	29.14	31.32	36.12
15	3.48	4.60	5.23	6.26	7.26	25.00	27.49	30.58	32.80	37.70
16	3.94	5.14	5.81	6.91	7.96	26.30	28.85	32.00	34.27	39.25
17	4.42	5.70	6.41	7.56	8.67	27.59	30.19	33.41	35.72	40.79
18	4.90	6.26	7.01	8.23	9.39	28.87	31.53	34.81	37.16	42.31
19	5.41	6.84	7.63	8.91	10.12	30.14	32.85	36.19	38.58	43.82
20	5.92	7.43	8.26	9.59	10.85	31.41	34.17	37.57	40.00	45.31
21	6.45	8.03	8.90	10.28	11.59	32.67	35.48	38.93	41.40	46.80
22	6.98	8.64	9.54	10.98	12.34	33.92	36.78	40.29	42.80	48.27
23	7.53	9.26	10.20	11.69	13.09	35.17	38.08	41.64	44.18	49.73
24	8.08	9.89	10.86	12.40	13.85	36.42	39.36	42.98	45.56	51.18
25	8.65	10.52	11.52	13.12	14.61	37.65	40.65	44.31	46.93	52.62
26	9.22	11.16	12.20	13.84	15.38	38.89	41.92	45.64	48.29	54.05
27	9.80	11.81	12.88	14.57	16.15	40.11	43.19	46.96	49.64	55.48
28	10.39	12.46	13.56	15.31	16.93	41.34	44.46	48.28	50.99	56.89
29	10.99	13.12	14.26	16.05	17.71	42.56	45.72	49.59	52.34	58.30
30	11.59	13.79	14.95	16.79	18.49	43.77	46.98	50.89	53.67	59.70
31	12.20	14.46	15.66	17.54	19.28	44.99	48.23	52.19	55.00	61.10
32	12.81	15.13	16.36	18.29	20.07	46.19	49.48	53.49	56.33	62.49
33	13.43	15.82	17.07	19.05	20.87	47.40	50.73	54.78	57.65	63.87
34	14.06	16.50	17.79	19.81	21.66	48.60	51.97	56.06	58.96	65.25
35	14.69	17.19	18.51	20.57	22.47	49.80	53.20	57.34	60.27	66.62
36	15.32	17.89	19.23	21.34	23.27	51.00	54.44	58.62	61.58	67.99
37	15.97	18.59	19.96	22.11	24.07	52.19	55.67	59.89	62.88	69.35
38	16.61	19.29	20.69	22.88	24.88	53.38	56.90	61.16	64.18	70.70
39	17.26	20.00	21.43	23.65	25.70	54.57	58.12	62.43	65.48	72.05
40	17.92	20.71	22.16	24.43	26.51	55.76	59.34	63.69	66.77	73.40
41	18.58	21.42	22.91	25.21	27.33	56.94	60.56	64.95	68.05	74.74
42	19.24	22.14	23.65	26.00	28.14	58.12	61.78	66.21	69.34	76.08
43	19.91	22.86	24.40	26.79	28.96	59.30	62.99	67.46	70.62	77.42
44	20.58	23.58	25.15	27.57	29.79	60.48	64.20	68.71	71.89	78.75
45	21.25	24.31	25.90	28.37	30.61	61.66	65.41	69.96	73.17	80.08
46	21.93	25.04	26.66	29.16	31.44	62.83	66.62	71.20	74.44	81.40
47	22.61	25.77	27.42	29.96	32.27	64.00	67.82	72.44	75.70	82.72
48	23.29	26.51	28.18	30.75	33.10	65.17	69.02	73.68	76.97	84.04
49	23.98	27.25	28.94	31.55	33.93	66.34	70.22	74.92	78.23	85.35
50	24.67	27.99	29.71	32.36	34.76	67.50	71.42	76.15	79.49	86.66

	α									
df	**0.999**	**0.995**	**0.99**	**0.975**	**0.95**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**
51	25.37	28.73	30.48	33.16	35.60	68.67	72.62	77.39	80.75	87.97
52	26.07	29.48	31.25	33.97	36.44	69.83	73.81	78.62	82.00	89.27
53	26.76	30.23	32.02	34.78	37.28	70.99	75.00	79.84	83.25	90.57
54	27.47	30.98	32.79	35.59	38.12	72.15	76.19	81.07	84.50	91.87
55	28.17	31.73	33.57	36.40	38.96	73.31	77.38	82.29	85.75	93.17
56	28.88	32.49	34.35	37.21	39.80	74.47	78.57	83.51	86.99	94.46
57	29.59	33.25	35.13	38.03	40.65	75.62	79.75	84.73	88.24	95.75
58	30.30	34.01	35.91	38.84	41.49	76.78	80.94	85.95	89.48	97.04
59	31.02	34.77	36.70	39.66	42.34	77.93	82.12	87.17	90.72	98.32
60	31.74	35.53	37.48	40.48	43.19	79.08	83.30	88.38	91.95	99.61
61	32.46	36.30	38.27	41.30	44.04	80.23	84.48	89.59	93.19	100.89
62	33.18	37.07	39.06	42.13	44.89	81.38	85.65	90.80	94.42	102.17
63	33.91	37.84	39.86	42.95	45.74	82.53	86.83	92.01	95.65	103.44
64	34.63	38.61	40.65	43.78	46.59	83.68	88.00	93.22	96.88	104.72
65	35.36	39.38	41.44	44.60	47.45	84.82	89.18	94.42	98.11	105.99
66	36.09	40.16	42.24	45.43	48.31	85.96	90.35	95.63	99.33	107.26
67	36.83	40.94	43.04	46.26	49.16	87.11	91.52	96.83	100.55	108.53
68	37.56	41.71	43.84	47.09	50.02	88.25	92.69	98.03	101.78	109.79
69	38.30	42.49	44.64	47.92	50.88	89.39	93.86	99.23	103.00	111.06
70	39.04	43.28	45.44	48.76	51.74	90.53	95.02	100.43	104.21	112.32
71	39.78	44.06	46.25	49.59	52.60	91.67	96.19	101.62	105.43	113.58
72	40.52	44.84	47.05	50.43	53.46	92.81	97.35	102.82	106.65	114.84
73	41.26	45.63	47.86	51.26	54.33	93.95	98.52	104.01	107.86	116.09
74	42.01	46.42	48.67	52.10	55.19	95.08	99.68	105.20	109.07	117.35
75	42.76	47.21	49.48	52.94	56.05	96.22	100.84	106.39	110.29	118.60
76	43.51	48.00	50.29	53.78	56.92	97.35	102.00	107.58	111.50	119.85
77	44.26	48.79	51.10	54.62	57.79	98.48	103.16	108.77	112.70	121.10
78	45.01	49.58	51.91	55.47	58.65	99.62	104.32	109.96	113.91	122.35
79	45.76	50.38	52.72	56.31	59.52	100.75	105.47	111.14	115.12	123.59
80	46.52	51.17	53.54	57.15	60.39	101.88	106.63	112.33	116.32	124.84
81	47.28	51.97	54.36	58.00	61.26	103.01	107.78	113.51	117.52	126.08
82	48.04	52.77	55.17	58.84	62.13	104.14	108.94	114.69	118.73	127.32
83	48.80	53.57	55.99	59.69	63.00	105.27	110.09	115.88	119.93	128.56
84	49.56	54.37	56.81	60.54	63.88	106.39	111.24	117.06	121.13	129.80
85	50.32	55.17	57.63	61.39	64.75	107.52	112.39	118.24	122.32	131.04
86	51.08	55.97	58.46	62.24	65.62	108.65	113.54	119.41	123.52	132.28
87	51.85	56.78	59.28	63.09	66.50	109.77	114.69	120.59	124.72	133.51
88	52.62	57.58	60.10	63.94	67.37	110.90	115.84	121.77	125.91	134.75
89	53.39	58.39	60.93	64.79	68.25	112.02	116.99	122.94	127.11	135.98
90	54.16	59.20	61.75	65.65	69.13	113.15	118.14	124.12	128.30	137.21
91	54.93	60.00	62.58	66.50	70.00	114.27	119.28	125.29	129.49	138.44
92	55.70	60.81	63.41	67.36	70.88	115.39	120.43	126.46	130.68	139.67
93	56.47	61.63	64.24	68.21	71.76	116.51	121.57	127.63	131.87	140.89
94	57.25	62.44	65.07	69.07	72.64	117.63	122.72	128.80	133.06	142.12
95	58.02	63.25	65.90	69.92	73.52	118.75	123.86	129.97	134.25	143.34
96	58.80	64.06	66.73	70.78	74.40	119.87	125.00	131.14	135.43	144.57
97	59.58	64.88	67.56	71.64	75.28	120.99	126.14	132.31	136.62	145.79
98	60.36	65.69	68.40	72.50	76.16	122.11	127.28	133.48	137.80	147.01
99	61.14	66.51	69.23	73.36	77.05	123.23	128.42	134.64	138.99	148.23
100	61.92	67.33	70.06	74.22	77.93	124.34	129.56	135.81	140.17	149.45

Statistical Table B: The standard normal (*Z*) distribution

This table gives probabilities under the right tail of the standard normal distribution. To determine the probability of sampling a value greater than or equal to a specific value of *Z*, find the first two digits of *Z* (called *a* and *b*) in the far left column of the table, and then find the last digit (*c*) in the top row of the table. For example, to find the probability of sampling a value greater than or equal to $Z = 1.96$, use $a.b = 1.9$ and $c = 6$. This probability is 0.025, or 2.5%.

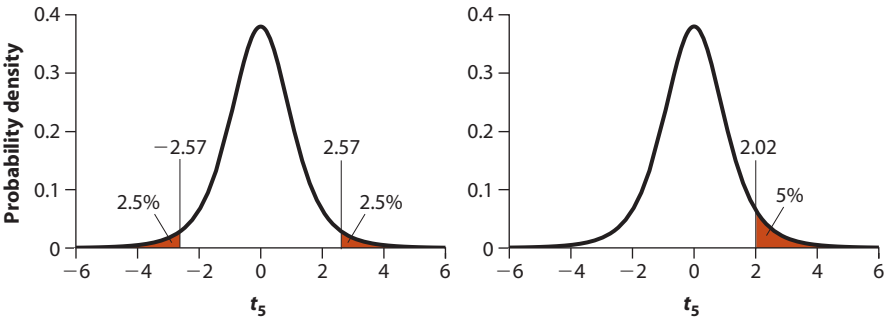


An example of a probability under the standard normal curve: the probability of sampling a value greater than or equal to the value 1.96 is 0.025, or 2.5%.

First two digits of *a.bc*	Second digit after decimal (*c*)									
	0	1	2	3	4	5	6	7	8	9
0.0	0.5	0.49601	0.49202	0.48803	0.48405	0.48006	0.47608	0.47210	0.46812	0.46414
0.1	0.46017	0.45620	0.45224	0.44828	0.44433	0.44038	0.43644	0.43251	0.42858	0.42465
0.2	0.42074	0.41683	0.41294	0.40905	0.40517	0.40129	0.39743	0.39358	0.38974	0.38591
0.3	0.38209	0.37828	0.37448	0.37070	0.36693	0.36317	0.35942	0.35569	0.35197	0.34827
0.4	0.34458	0.34090	0.33724	0.33360	0.32997	0.32636	0.32276	0.31918	0.31561	0.31207
0.5	0.30854	0.30503	0.30153	0.29806	0.29460	0.29116	0.28774	0.28434	0.28096	0.27760
0.6	0.27425	0.27093	0.26763	0.26435	0.26109	0.25785	0.25463	0.25143	0.24825	0.24510
0.7	0.24196	0.23885	0.23576	0.23270	0.22965	0.22663	0.22363	0.22065	0.21770	0.21476
0.8	0.21186	0.20897	0.20611	0.20327	0.20045	0.19766	0.19489	0.19215	0.18943	0.18673
0.9	0.18406	0.18141	0.17879	0.17619	0.17361	0.17106	0.16853	0.16602	0.16354	0.16109
1.0	0.15866	0.15625	0.15386	0.15151	0.14917	0.14686	0.14457	0.14231	0.14007	0.13786
1.1	0.13567	0.13350	0.13136	0.12924	0.12714	0.12507	0.12302	0.12100	0.11900	0.11702
1.2	0.11507	0.11314	0.11123	0.10935	0.10749	0.10565	0.10383	0.10204	0.10027	0.09853
1.3	0.09680	0.09510	0.09342	0.09176	0.09012	0.08851	0.08691	0.08534	0.08379	0.08226
1.4	0.08076	0.07927	0.07780	0.07636	0.07493	0.07353	0.07215	0.07078	0.06944	0.06811
1.5	0.06681	0.06552	0.06426	0.06301	0.06178	0.06057	0.05938	0.05821	0.05705	0.05592
1.6	0.05480	0.05370	0.05262	0.05155	0.05050	0.04947	0.04846	0.04746	0.04648	0.04551
1.7	0.04457	0.04363	0.04272	0.04182	0.04093	0.04006	0.03920	0.03836	0.03754	0.03673
1.8	0.03593	0.03515	0.03438	0.03362	0.03288	0.03216	0.03144	0.03074	0.03005	0.02938
1.9	0.02872	0.02807	0.02743	0.02680	0.02619	0.02559	0.02500	0.02442	0.02385	0.02330
2.0	0.02275	0.02222	0.02169	0.02118	0.02068	0.02018	0.01970	0.01923	0.01876	0.01831
2.1	0.01786	0.01743	0.01700	0.01659	0.01618	0.01578	0.01539	0.01500	0.01463	0.01426
2.2	0.01390	0.01355	0.01321	0.01287	0.01255	0.01222	0.01191	0.01160	0.01130	0.01101
2.3	0.01072	0.01044	0.01017	0.00990	0.00964	0.00939	0.00914	0.00889	0.00866	0.00842
2.4	0.00820	0.00798	0.00776	0.00755	0.00734	0.00714	0.00695	0.00676	0.00657	0.00639
2.5	0.00621	0.00604	0.00587	0.00570	0.00554	0.00539	0.00523	0.00508	0.00494	0.00480
2.6	0.00466	0.00453	0.00440	0.00427	0.00415	0.00402	0.00391	0.00379	0.00368	0.00357
2.7	0.00347	0.00336	0.00326	0.00317	0.00307	0.00298	0.00289	0.00280	0.00272	0.00264
2.8	0.00256	0.00248	0.00240	0.00233	0.00226	0.00219	0.00212	0.00205	0.00199	0.00193
2.9	0.00187	0.00181	0.00175	0.00169	0.00164	0.00159	0.00154	0.00149	0.00144	0.00139
3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00104	0.00100
3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
4.0	0.00003	0.00003	0.00003	0.00003	0.00003	0.00003	0.00002	0.00002	0.00002	0.00002

Statistical Table C: Student's *t*-distribution

This table gives critical values of the *t*-distribution. The two-tailed critical value $t_{\alpha(2),df}$ defines the combined area α under both tails of the *t*-distribution having *df* degrees of freedom. To find a critical value in the table, select the desired value of $\alpha(2)$ along the top and the number of degrees of freedom in the far left column. For example, if $df = 5$ and $\alpha(2) = 0.05$, the critical value is 2.57; that is, the probability of a *t*-value greater than or equal to 2.57, or less than or equal to -2.57, is 0.05. The left panel of the following figure illustrates the critical value for this example.



Critical values of the *t*-distribution having five degrees of freedom when $\alpha = 0.05$. The left panel shows the two-tailed case. The area in red shows the combined tail probabilities of 0.05, with 2.5% of the probability in each tail. The boundaries of the areas in red are -2.57 and 2.57. The right panel shows the one-tailed case. The area in red indicates the right tail of the distribution corresponding to 5% of the total probability. The boundary of the red section is 2.02.

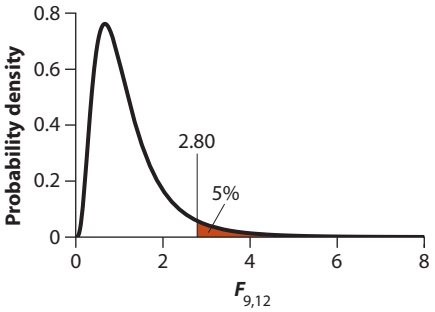
The one-tailed critical value $t_{\alpha(1),df}$ defines the area α under the right tail of the *t*-distribution having *df* degrees of freedom. To find a critical value in the table, select the desired value of $\alpha(1)$ along the top and the number of degrees of freedom in the far left column. For example, if $df = 5$ and $\alpha(1) = 0.05$, the critical value is 2.02; that is, the probability of a value greater than or equal to 2.02 is 0.05. The right panel of the preceding figure illustrates the critical value for this example.

	$\alpha(2)$:	0.2	0.10	0.05	0.02	0.01	0.001	0.0001
df	$\alpha(1)$:	**0.1**	**0.05**	**0.025**	**0.01**	**0.005**	**0.0005**	**0.00005**
1		3.08	6.31	12.71	31.82	63.66	636.62	6366.20
2		1.89	2.92	4.30	6.96	9.92	31.60	99.99
3		1.64	2.35	3.18	4.54	5.84	12.92	28.00
4		1.53	2.13	2.78	3.75	4.60	8.61	15.54
5		1.48	2.02	2.57	3.36	4.03	6.87	11.18
6		1.44	1.94	2.45	3.14	3.71	5.96	9.08
7		1.41	1.89	2.36	3.00	3.50	5.41	7.88
8		1.40	1.86	2.31	2.90	3.36	5.04	7.12
9		1.38	1.83	2.26	2.82	3.25	4.78	6.59
10		1.37	1.81	2.23	2.76	3.17	4.59	6.21
11		1.36	1.80	2.20	2.72	3.11	4.44	5.92
12		1.36	1.78	2.18	2.68	3.05	4.32	5.69
13		1.35	1.77	2.16	2.65	3.01	4.22	5.51
14		1.35	1.76	2.14	2.62	2.98	4.14	5.36
15		1.34	1.75	2.13	2.60	2.95	4.07	5.24
16		1.34	1.75	2.12	2.58	2.92	4.01	5.13
17		1.33	1.74	2.11	2.57	2.90	3.97	5.04
18		1.33	1.73	2.10	2.55	2.88	3.92	4.97
19		1.33	1.73	2.09	2.54	2.86	3.88	4.90
20		1.33	1.72	2.09	2.53	2.85	3.85	4.84
21		1.32	1.72	2.08	2.52	2.83	3.82	4.78
22		1.32	1.72	2.07	2.51	2.82	3.79	4.74
23		1.32	1.71	2.07	2.50	2.81	3.77	4.69
24		1.32	1.71	2.06	2.49	2.80	3.75	4.65
25		1.32	1.71	2.06	2.49	2.79	3.73	4.62
26		1.31	1.71	2.06	2.48	2.78	3.71	4.59
27		1.31	1.70	2.05	2.47	2.77	3.69	4.56
28		1.31	1.70	2.05	2.47	2.76	3.67	4.53
29		1.31	1.70	2.05	2.46	2.76	3.66	4.51
30		1.31	1.70	2.04	2.46	2.75	3.65	4.48
31		1.31	1.70	2.04	2.45	2.74	3.63	4.46
32		1.31	1.69	2.04	2.45	2.74	3.62	4.44
33		1.31	1.69	2.03	2.44	2.73	3.61	4.42
34		1.31	1.69	2.03	2.44	2.73	3.60	4.41
35		1.31	1.69	2.03	2.44	2.72	3.59	4.39
36		1.31	1.69	2.03	2.43	2.72	3.58	4.37
37		1.30	1.69	2.03	2.43	2.72	3.57	4.36
38		1.30	1.69	2.02	2.43	2.71	3.57	4.35
39		1.30	1.68	2.02	2.43	2.71	3.56	4.33
40		1.30	1.68	2.02	2.42	2.70	3.55	4.32
41		1.30	1.68	2.02	2.42	2.70	3.54	4.31
42		1.30	1.68	2.02	2.42	2.70	3.54	4.30
43		1.30	1.68	2.02	2.42	2.70	3.53	4.29
44		1.30	1.68	2.02	2.41	2.69	3.53	4.28
45		1.30	1.68	2.01	2.41	2.69	3.52	4.27
46		1.30	1.68	2.01	2.41	2.69	3.51	4.26
47		1.30	1.68	2.01	2.41	2.68	3.51	4.25
48		1.30	1.68	2.01	2.41	2.68	3.51	4.24
49		1.30	1.68	2.01	2.40	2.68	3.50	4.24
50		1.30	1.68	2.01	2.40	2.68	3.50	4.23

	α(2):	0.2	0.10	0.05	0.02	0.01	0.001	0.0001
df	α(1):	0.1	0.05	0.025	0.01	0.005	0.0005	0.00005
51		1.30	1.68	2.01	2.40	2.68	3.49	4.22
52		1.30	1.67	2.01	2.40	2.67	3.49	4.21
53		1.30	1.67	2.01	2.40	2.67	3.48	4.21
54		1.30	1.67	2.00	2.40	2.67	3.48	4.20
55		1.30	1.67	2.00	2.40	2.67	3.48	4.20
56		1.30	1.67	2.00	2.39	2.67	3.47	4.19
57		1.30	1.67	2.00	2.39	2.66	3.47	4.18
58		1.30	1.67	2.00	2.39	2.66	3.47	4.18
59		1.30	1.67	2.00	2.39	2.66	3.46	4.17
60		1.30	1.67	2.00	2.39	2.66	3.46	4.17
61		1.30	1.67	2.00	2.39	2.66	3.46	4.16
62		1.30	1.67	2.00	2.39	2.66	3.45	4.16
63		1.30	1.67	2.00	2.39	2.66	3.45	4.15
64		1.29	1.67	2.00	2.39	2.65	3.45	4.15
65		1.29	1.67	2.00	2.39	2.65	3.45	4.15
66		1.29	1.67	2.00	2.38	2.65	3.44	4.14
67		1.29	1.67	2.00	2.38	2.65	3.44	4.14
68		1.29	1.67	2.00	2.38	2.65	3.44	4.13
69		1.29	1.67	1.99	2.38	2.65	3.44	4.13
70		1.29	1.67	1.99	2.38	2.65	3.44	4.13
71		1.29	1.67	1.99	2.38	2.65	3.43	4.12
72		1.29	1.67	1.99	2.38	2.65	3.43	4.12
73		1.29	1.67	1.99	2.38	2.64	3.43	4.12
74		1.29	1.67	1.99	2.38	2.64	3.43	4.11
75		1.29	1.67	1.99	2.38	2.64	3.43	4.11
76		1.29	1.67	1.99	2.38	2.64	3.42	4.11
77		1.29	1.66	1.99	2.38	2.64	3.42	4.10
78		1.29	1.66	1.99	2.38	2.64	3.42	4.10
79		1.29	1.66	1.99	2.37	2.64	3.42	4.10
80		1.29	1.66	1.99	2.37	2.64	3.42	4.10
81		1.29	1.66	1.99	2.37	2.64	3.41	4.09
82		1.29	1.66	1.99	2.37	2.64	3.41	4.09
83		1.29	1.66	1.99	2.37	2.64	3.41	4.09
84		1.29	1.66	1.99	2.37	2.64	3.41	4.09
85		1.29	1.66	1.99	2.37	2.63	3.41	4.08
86		1.29	1.66	1.99	2.37	2.63	3.41	4.08
87		1.29	1.66	1.99	2.37	2.63	3.41	4.08
88		1.29	1.66	1.99	2.37	2.63	3.40	4.08
89		1.29	1.66	1.99	2.37	2.63	3.40	4.07
90		1.29	1.66	1.99	2.37	2.63	3.40	4.07
100		1.29	1.66	1.98	2.36	2.63	3.39	4.05
120		1.29	1.66	1.98	2.36	2.62	3.37	4.03
140		1.29	1.66	1.98	2.35	2.61	3.36	4.01
160		1.29	1.65	1.97	2.35	2.61	3.35	3.99
180		1.29	1.65	1.97	2.35	2.60	3.35	3.98
200		1.29	1.65	1.97	2.35	2.60	3.34	3.97
400		1.28	1.65	1.97	2.34	2.59	3.32	3.93
1000		1.28	1.65	1.96	2.33	2.58	3.30	3.91

Statistical Table D: The *F*-distribution

These tables give critical values of the *F*-distribution for $\alpha(1) = 0.05$ and $\alpha(1) = 0.01$. The critical value $F_{\alpha(1),df1,df2}$ defines the area α under the right tail of the *F*-distribution having df_1 and df_2 degrees of freedom. To find a critical value in the table, select the numerator degrees of freedom (df_1) listed across the top row and the denominator degrees of freedom (df_2) given in the first column. For example, if $df_1 = 9$, $df_2 = 12$, and $\alpha = 0.05$, then the critical value is 2.80; that is, the probability of a value greater than or equal to 2.80 is 0.05.



The critical value for the *F*-distribution having nine and 12 degrees of freedom when $\alpha = 0.05$. The area in red indicates the tail probability corresponding to 0.05, or 5%. The boundary of the red section is 2.80.

Critical value of F, $\alpha(1) = 0.05$

Denominator *df*	Numerator *df*									
	1	2	3	4	5	6	7	8	9	10
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13	2.07	2.03
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99
70	3.98	3.13	2.74	2.50	2.35	2.23	2.14	2.07	2.02	1.97
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95
90	3.95	3.10	2.71	2.47	2.32	2.20	2.11	2.04	1.99	1.94
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93
200	3.89	3.04	2.65	2.42	2.26	2.14	2.06	1.98	1.93	1.88
400	3.86	3.02	2.63	2.39	2.24	2.12	2.03	1.96	1.90	1.85

Critical value of F, $\alpha(1) = 0.05$ (continued)

Denominator *df*	Numerator *df*									
	12	15	20	30	40	60	100	200	400	1000
1	243.91	245.95	248.01	250.10	251.14	252.20	253.04	253.68	254.00	254.19
2	19.41	19.43	19.45	19.46	19.47	19.48	19.49	19.49	19.49	19.49
3	8.74	8.70	8.66	8.62	8.59	8.57	8.55	8.54	8.53	8.53
4	5.91	5.86	5.80	5.75	5.72	5.69	5.66	5.65	5.64	5.63
5	4.68	4.62	4.56	4.50	4.46	4.43	4.41	4.39	4.38	4.37
6	4.00	3.94	3.87	3.81	3.77	3.74	3.71	3.69	3.68	3.67
7	3.57	3.51	3.44	3.38	3.34	3.30	3.27	3.25	3.24	3.23
8	3.28	3.22	3.15	3.08	3.04	3.01	2.97	2.95	2.94	2.93
9	3.07	3.01	2.94	2.86	2.83	2.79	2.76	2.73	2.72	2.71
10	2.91	2.85	2.77	2.70	2.66	2.62	2.59	2.56	2.55	2.54
11	2.79	2.72	2.65	2.57	2.53	2.49	2.46	2.43	2.42	2.41
12	2.69	2.62	2.54	2.47	2.43	2.38	2.35	2.32	2.31	2.30
13	2.60	2.53	2.46	2.38	2.34	2.30	2.26	2.23	2.22	2.21
14	2.53	2.46	2.39	2.31	2.27	2.22	2.19	2.16	2.15	2.14
15	2.48	2.40	2.33	2.25	2.20	2.16	2.12	2.10	2.08	2.07
16	2.42	2.35	2.28	2.19	2.15	2.11	2.07	2.04	2.02	2.02
17	2.38	2.31	2.23	2.15	2.10	2.06	2.02	1.99	1.98	1.97
18	2.34	2.27	2.19	2.11	2.06	2.02	1.98	1.95	1.93	1.92
19	2.31	2.23	2.16	2.07	2.03	1.98	1.94	1.91	1.89	1.88
20	2.28	2.20	2.12	2.04	1.99	1.95	1.91	1.88	1.86	1.85
21	2.25	2.18	2.10	2.01	1.96	1.92	1.88	1.84	1.83	1.82
22	2.23	2.15	2.07	1.98	1.94	1.89	1.85	1.82	1.80	1.79
23	2.20	2.13	2.05	1.96	1.91	1.86	1.82	1.79	1.77	1.76
24	2.18	2.11	2.03	1.94	1.89	1.84	1.80	1.77	1.75	1.74
25	2.16	2.09	2.01	1.92	1.87	1.82	1.78	1.75	1.73	1.72
26	2.15	2.07	1.99	1.90	1.85	1.80	1.76	1.73	1.71	1.70
27	2.13	2.06	1.97	1.88	1.84	1.79	1.74	1.71	1.69	1.68
28	2.12	2.04	1.96	1.87	1.82	1.77	1.73	1.69	1.67	1.66
29	2.10	2.03	1.94	1.85	1.81	1.75	1.71	1.67	1.66	1.65
30	2.09	2.01	1.93	1.84	1.79	1.74	1.70	1.66	1.64	1.63
40	2.00	1.92	1.84	1.74	1.69	1.64	1.59	1.55	1.53	1.52
50	1.95	1.87	1.78	1.69	1.63	1.58	1.52	1.48	1.46	1.45
60	1.92	1.84	1.75	1.65	1.59	1.53	1.48	1.44	1.41	1.40
70	1.89	1.81	1.72	1.62	1.57	1.50	1.45	1.40	1.38	1.36
80	1.88	1.79	1.70	1.60	1.54	1.48	1.43	1.38	1.35	1.34
90	1.86	1.78	1.69	1.59	1.53	1.46	1.41	1.36	1.33	1.31
100	1.85	1.77	1.68	1.57	1.52	1.45	1.39	1.34	1.31	1.30
200	1.80	1.72	1.62	1.52	1.46	1.39	1.32	1.26	1.23	1.21
400	1.78	1.69	1.60	1.49	1.42	1.35	1.28	1.22	1.18	1.15

Critical value of F, $\alpha(1) = 0.01$

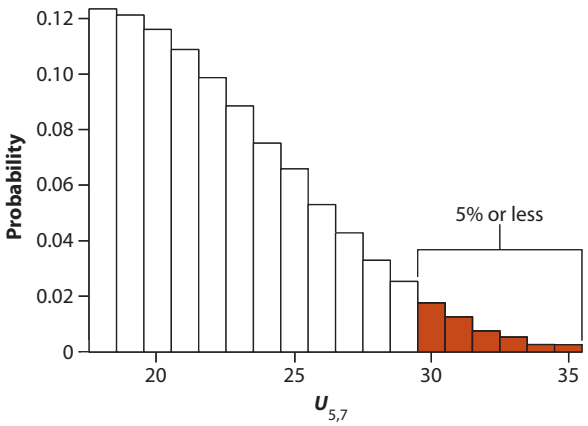
Denominator *df*	Numerator *df*									
	1	2	3	4	5	6	7	8	9	10
1	4052	4999	5403	5624	5763	5859	5928	5981	6022	6055
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80
50	7.17	5.06	4.20	3.72	3.41	3.19	3.02	2.89	2.78	2.70
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63
70	7.01	4.92	4.07	3.60	3.29	3.07	2.91	2.78	2.67	2.59
80	6.96	4.88	4.04	3.56	3.26	3.04	2.87	2.74	2.64	2.55
90	6.93	4.85	4.01	3.53	3.23	3.01	2.84	2.72	2.61	2.52
100	6.90	4.82	3.98	3.51	3.21	2.99	2.82	2.69	2.59	2.50
200	6.76	4.71	3.88	3.41	3.11	2.89	2.73	2.60	2.50	2.41
400	6.70	4.66	3.83	3.37	3.06	2.85	2.68	2.56	2.45	2.37

Critical value of F, $\alpha(1) = 0.01$ (continued)

Denominator *df*	Numerator *df*									
	12	**15**	**20**	**30**	**40**	**60**	**100**	**200**	**400**	**1000**
1	6106	6157	6208	6260	6286	6313	6334	6350	6357	6362
2	99.42	99.43	99.45	99.47	99.47	99.48	99.49	99.49	99.50	99.50
3	27.05	26.87	26.69	26.50	26.41	26.32	26.24	26.18	26.15	26.14
4	14.37	14.20	14.02	13.84	13.75	13.65	13.58	13.52	13.49	13.47
5	9.89	9.72	9.55	9.38	9.29	9.20	9.13	9.08	9.05	9.03
6	7.72	7.56	7.40	7.23	7.14	7.06	6.99	6.93	6.91	6.89
7	6.47	6.31	6.16	5.99	5.91	5.82	5.75	5.70	5.68	5.66
8	5.67	5.52	5.36	5.20	5.12	5.03	4.96	4.91	4.89	4.87
9	5.11	4.96	4.81	4.65	4.57	4.48	4.41	4.36	4.34	4.32
10	4.71	4.56	4.41	4.25	4.17	4.08	4.01	3.96	3.94	3.92
11	4.40	4.25	4.10	3.94	3.86	3.78	3.71	3.66	3.63	3.61
12	4.16	4.01	3.86	3.70	3.62	3.54	3.47	3.41	3.39	3.37
13	3.96	3.82	3.66	3.51	3.43	3.34	3.27	3.22	3.19	3.18
14	3.80	3.66	3.51	3.35	3.27	3.18	3.11	3.06	3.03	3.02
15	3.67	3.52	3.37	3.21	3.13	3.05	2.98	2.92	2.90	2.88
16	3.55	3.41	3.26	3.10	3.02	2.93	2.86	2.81	2.78	2.76
17	3.46	3.31	3.16	3.00	2.92	2.83	2.76	2.71	2.68	2.66
18	3.37	3.23	3.08	2.92	2.84	2.75	2.68	2.62	2.59	2.58
19	3.30	3.15	3.00	2.84	2.76	2.67	2.60	2.55	2.52	2.50
20	3.23	3.09	2.94	2.78	2.69	2.61	2.54	2.48	2.45	2.43
21	3.17	3.03	2.88	2.72	2.64	2.55	2.48	2.42	2.39	2.37
22	3.12	2.98	2.83	2.67	2.58	2.50	2.42	2.36	2.34	2.32
23	3.07	2.93	2.78	2.62	2.54	2.45	2.37	2.32	2.29	2.27
24	3.03	2.89	2.74	2.58	2.49	2.40	2.33	2.27	2.24	2.22
25	2.99	2.85	2.70	2.54	2.45	2.36	2.29	2.23	2.20	2.18
26	2.96	2.81	2.66	2.50	2.42	2.33	2.25	2.19	2.16	2.14
27	2.93	2.78	2.63	2.47	2.38	2.29	2.22	2.16	2.13	2.11
28	2.90	2.75	2.60	2.44	2.35	2.26	2.19	2.13	2.10	2.08
29	2.87	2.73	2.57	2.41	2.33	2.23	2.16	2.10	2.07	2.05
30	2.84	2.70	2.55	2.39	2.30	2.21	2.13	2.07	2.04	2.02
40	2.66	2.52	2.37	2.20	2.11	2.02	1.94	1.87	1.84	1.82
50	2.56	2.42	2.27	2.10	2.01	1.91	1.82	1.76	1.72	1.70
60	2.50	2.35	2.20	2.03	1.94	1.84	1.75	1.68	1.64	1.62
70	2.45	2.31	2.15	1.98	1.89	1.78	1.70	1.62	1.58	1.56
80	2.42	2.27	2.12	1.94	1.85	1.75	1.65	1.58	1.54	1.51
90	2.39	2.24	2.09	1.92	1.82	1.72	1.62	1.55	1.50	1.48
100	2.37	2.22	2.07	1.89	1.80	1.69	1.60	1.52	1.47	1.45
200	2.27	2.13	1.97	1.79	1.69	1.58	1.48	1.39	1.34	1.30
400	2.23	2.08	1.92	1.75	1.64	1.53	1.42	1.32	1.26	1.2

Statistical Table E: Mann–Whitney *U*-distribution

These tables give the two-tailed critical values of the *U*-distribution for $\alpha = 0.05$ and $\alpha = 0.01$. *U* is the larger of U_1 and U_2. The critical value $U_{0.05(2),n1,n2}$ defines the area α under the right tail of the *U*-distribution corresponding to sample sizes n_1 and n_2. To find a critical value in the table, select n_1 from the top row and n_2 from the far left column. A "—" means that it is not possible to reject a null hypothesis with that α and those sample sizes. For larger sample sizes, use the *Z*-approximation from Chapter 13. For example, if $n_1 = 5$, $n_2 = 7$, and $\alpha = 0.05$, then the critical value is 30; that is, the probability of a value greater than or equal to 30 is 0.05 or less (it may be less than 0.05 because the *U*-distribution is discrete, and no critical value may correspond exactly to 0.05). The following figure illustrates the critical value for this example.



The two-tailed critical value of the Mann–Whitney *U*-distribution when $n_1 = 5, n_2 = 7$, and $\alpha = 0.05$. The area in red indicates the tail probability corresponding to 0.05, or 5%. The boundary of the red section is 30. Thus, the probability of a value greater than or equal to 30 is 0.05 or less. (The distribution gives the two-tailed value, because it measures the probability that either U_1 or U_2 is greater than $U_{0.05(2),n1,n2}$.)

$\alpha(2) = 0.05$

	n_1												
n_2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	—	—	15	17	20	22	25	27	30	32	35	37	40
4	—	16	19	22	25	28	32	35	38	41	44	47	50
5	15	19	23	27	30	34	38	42	46	49	53	57	61
6	17	22	27	31	36	40	44	49	53	58	62	67	71
7	20	25	30	36	41	46	51	56	61	66	71	76	81
8	22	28	34	40	46	51	57	63	69	74	80	86	91
9	25	32	38	44	51	57	64	70	76	82	89	95	101
10	27	35	42	49	56	63	70	77	84	91	97	104	111
11	30	38	46	53	61	69	76	84	91	99	106	114	121
12	32	41	49	58	66	74	82	91	99	107	115	123	131
13	35	44	53	62	71	80	89	97	106	115	124	132	141
14	37	47	57	67	76	86	95	104	114	123	132	141	151
15	40	50	61	71	81	91	101	111	121	131	141	151	161

$\alpha(2) = 0.01$

	n_1												
n_2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	—	—	—	—	—	—	27	30	33	35	38	41	43
4	—	—	—	24	28	31	35	38	42	45	49	52	55
5	—	—	25	29	34	38	42	46	50	54	58	63	67
6	—	24	29	34	39	44	49	54	59	63	68	73	78
7	—	28	34	39	45	50	56	61	67	72	78	83	89
8	—	31	38	44	50	57	63	69	75	81	87	94	100
9	27	35	42	49	56	63	70	77	83	90	97	104	111
10	30	38	46	54	61	69	77	84	92	99	106	114	121
11	33	42	50	59	67	75	83	92	100	108	116	124	132
12	35	45	54	63	72	81	90	99	108	117	125	134	143
13	38	49	58	68	78	87	97	106	116	125	135	144	153
14	41	52	63	73	83	94	104	114	124	134	144	154	164
15	43	53	67	78	89	100	111	121	132	143	153	164	174

Statistical Table F: Tukey–Kramer *q*-distribution

This table gives the critical values of the Tukey–Kramer q-distribution for $\alpha = 0.05$. The critical value $q_{0.05,k,N-k}$ defines the area 0.05 under the right tail of the q-distribution having k groups and $N - k$ degrees of freedom, where N is the total sample size of all groups combined. The probability of a q-value greater than or equal to $q_{0.05,k,N-k}$ is 0.05, or 5%.

	Number of groups (*k*)													
df_{error}	2	3	4	5	6	7	8	9	10	11	12	13	14	15
10	2.23	2.74	3.06	3.29	3.47	3.62	3.75	3.86	3.96	4.05	4.12	4.20	4.26	4.32
11	2.20	2.70	3.01	3.23	3.41	3.56	3.68	3.79	3.88	3.96	4.04	4.11	4.17	4.23
12	2.18	2.67	2.97	3.19	3.36	3.50	3.62	3.72	3.81	3.90	3.97	4.04	4.10	4.16
13	2.16	2.64	2.94	3.15	3.32	3.45	3.57	3.67	3.76	3.84	3.91	3.98	4.04	4.09
14	2.14	2.62	2.91	3.12	3.28	3.41	3.53	3.63	3.72	3.79	3.86	3.93	3.99	4.04
15	2.13	2.60	2.88	3.09	3.25	3.38	3.49	3.59	3.68	3.75	3.82	3.88	3.94	3.99
16	2.12	2.58	2.86	3.06	3.22	3.35	3.46	3.56	3.64	3.72	3.78	3.85	3.90	3.95
17	2.11	2.57	2.84	3.04	3.20	3.33	3.44	3.53	3.61	3.69	3.75	3.81	3.87	3.92
18	2.10	2.55	2.83	3.02	3.18	3.30	3.41	3.50	3.59	3.66	3.72	3.78	3.84	3.89
19	2.09	2.54	2.81	3.01	3.16	3.16	3.39	3.48	3.56	3.63	3.70	3.76	3.81	3.86
20	2.09	2.53	2.80	2.99	3.14	3.27	3.37	3.46	3.54	3.61	3.68	3.73	3.79	3.84
21	2.08	2.52	2.79	2.98	3.13	3.25	3.35	3.44	3.52	3.59	3.66	3.71	3.77	3.82
22	2.07	2.51	2.78	2.97	3.12	3.24	3.34	3.43	3.51	3.57	3.64	3.69	3.75	3.80
23	2.07	2.50	2.77	2.96	3.10	3.22	3.32	3.41	3.49	3.56	3.62	3.68	3.73	3.78
24	2.06	2.50	2.76	2.95	3.09	3.21	3.31	3.40	3.48	3.54	3.61	3.66	3.71	3.76
25	2.06	2.49	2.75	2.94	3.08	3.20	3.30	3.39	3.46	3.53	3.59	3.65	3.70	3.75
26	2.06	2.48	2.74	2.93	3.07	3.19	3.29	3.38	3.45	3.52	3.58	3.63	3.68	3.73
27	2.05	2.48	2.74	2.92	3.06	3.18	3.28	3.36	3.44	3.51	3.57	3.62	3.67	3.72
28	2.05	2.47	2.73	2.91	3.06	3.17	3.27	3.35	3.43	3.50	3.56	3.61	3.66	3.71
29	2.05	2.47	2.72	2.91	3.05	3.16	3.26	3.35	3.42	3.49	3.55	3.60	3.65	3.70
30	2.04	2.46	2.72	2.90	3.04	3.16	3.25	3.34	3.41	3.48	3.54	3.59	3.64	3.68
31	2.04	2.46	2.71	2.89	3.04	3.15	3.25	3.33	3.40	3.47	3.53	3.58	3.63	3.68
32	2.04	2.46	2.71	2.89	3.03	3.14	3.24	3.32	3.40	3.46	3.52	3.57	3.62	3.67
33	2.03	2.45	2.70	2.88	3.02	3.14	3.23	3.32	3.39	3.45	3.51	3.56	3.61	3.66
34	2.03	2.45	2.70	2.88	3.02	3.13	3.23	3.31	3.38	3.45	3.50	3.56	3.60	3.65
35	2.03	2.45	2.70	2.88	3.01	3.13	3.22	3.30	3.37	3.44	3.50	3.55	3.60	3.64
36	2.03	2.44	2.69	2.87	3.01	3.12	3.22	3.30	3.37	3.43	3.49	3.54	3.59	3.64
37	2.03	2.44	2.69	2.87	3.00	3.12	3.21	3.29	3.36	3.43	3.48	3.54	3.58	3.63
38	2.02	2.44	2.69	2.86	3.00	3.11	3.21	3.29	3.36	3.42	3.48	3.53	3.58	3.62
39	2.02	2.44	2.68	2.86	3.00	3.11	3.20	3.28	3.35	3.42	3.47	3.52	3.57	3.62
40	2.02	2.43	2.68	2.86	2.99	3.10	3.20	3.28	3.35	3.41	3.47	3.52	3.57	3.61
41	2.02	2.43	2.68	2.85	2.99	3.10	3.19	3.27	3.34	3.41	3.46	3.51	3.56	3.60
42	2.02	2.43	2.67	2.85	2.99	3.10	3.19	3.27	3.34	3.40	3.46	3.51	3.56	3.60
43	2.02	2.43	2.67	2.85	2.98	3.09	3.18	3.26	3.33	3.40	3.45	3.50	3.55	3.59
44	2.02	2.43	2.67	2.84	2.98	3.09	3.18	3.26	3.33	3.39	3.45	3.50	3.55	3.59
45	2.01	2.42	2.67	2.84	2.98	3.09	3.18	3.26	3.33	3.39	3.44	3.50	3.54	3.59
46	2.01	2.42	2.67	2.84	2.97	3.08	3.17	3.25	3.32	3.39	3.44	3.49	3.54	3.58
47	2.01	2.42	2.66	2.84	2.97	3.08	3.17	3.25	3.32	3.38	3.44	3.49	3.53	3.58
48	2.01	2.42	2.66	2.83	2.97	3.08	3.17	3.25	3.32	3.38	3.43	3.48	3.53	3.57
49	2.01	2.42	2.66	2.83	2.97	3.07	3.17	3.24	3.31	3.37	3.43	3.48	3.53	3.57
50	2.01	2.42	2.66	2.83	2.96	3.07	3.16	3.24	3.31	3.37	3.43	3.48	3.52	3.57

Statistical Table G: Critical values for the Spearman's rank correlation

This table gives two-tailed critical values of the Spearman rank correlation under the null hypothesis that the population correlation is zero. The critical value $r_{S(\alpha,n)}$ defines the combined area α under both tails of the null distribution, where n is the sample size. The probability of a value greater than or equal to $r_{S\,(\alpha,n)}$ or less than or equal to $-r_{S\,(\alpha,n)}$ is α. These critical values were obtained using the *SuppDist* package (Wheeler 2005) implemented in *R*, according to the methods of Kendall and Smith (1939).

n	$\alpha = 0.05$	$\alpha = 0.01$
5	0.900	
6	0.943	1.000
7	0.821	0.929
8	0.762	0.881
9	0.700	0.833
10	0.648	0.782
11	0.618	0.755
12	0.587	0.720
13	0.560	0.692
14	0.538	0.670
15	0.521	0.645
16	0.503	0.626
17	0.485	0.610
18	0.472	0.593
19	0.458	0.579
20	0.447	0.564
21	0.435	0.551
22	0.425	0.539
23	0.415	0.528
24	0.406	0.516
25	0.398	0.506
26	0.389	0.497
27	0.382	0.488
28	0.375	0.479
29	0.368	0.471
30	0.362	0.464
31	0.356	0.456
32	0.350	0.449
33	0.345	0.443
34	0.339	0.436
35	0.334	0.430
36	0.329	0.424
37	0.325	0.419
38	0.320	0.413
39	0.316	0.408
40	0.312	0.403
41	0.308	0.398
42	0.305	0.393
43	0.301	0.389
44	0.298	0.385
45	0.294	0.380
46	0.291	0.376
47	0.288	0.372
48	0.285	0.369
49	0.282	0.365
50	0.279	0.361
51	0.276	0.358
52	0.273	0.354
53	0.271	0.351
54	0.268	0.348
55	0.266	0.345
56	0.263	0.342
57	0.261	0.339
58	0.259	0.336
59	0.256	0.333
60	0.254	0.330
61	0.252	0.327
62	0.250	0.325
63	0.248	0.322
64	0.246	0.320
65	0.244	0.317
66	0.242	0.315
67	0.241	0.313
68	0.239	0.310
69	0.237	0.308
70	0.235	0.306
71	0.234	0.304
72	0.232	0.302
73	0.230	0.300
74	0.229	0.298
75	0.227	0.296
76	0.226	0.294
77	0.224	0.292
78	0.223	0.290
79	0.221	0.288
80	0.220	0.286
81	0.219	0.285
82	0.217	0.283
83	0.216	0.281
84	0.215	0.280
85	0.213	0.278
86	0.212	0.276
87	0.211	0.275
88	0.210	0.273
89	0.208	0.272
90	0.207	0.270
91	0.206	0.269
92	0.205	0.267
93	0.204	0.266
94	0.203	0.264
95	0.202	0.263
96	0.201	0.262
97	0.200	0.260
98	0.199	0.259
99	0.198	0.258
100	0.197	0.257